Mat 1575 HW 4 (§1.5)

In the following exercises, verify each identity using differentiation. Then, using the indicated u-substitution, identify f such that the integral takes the form $\int f(u)\,du$.

256. $\int x\sqrt{x+1}\,dx = \frac{2}{15}(x+1)^{3/2}(3x-2)+C; u=x+1$

Check by using differentiation:

$$\left(\frac{2}{15}(x+1)^{\frac{3}{2}}(3x-2)+C\right)' = \frac{3}{2}\cdot\frac{2}{15}(x+1)^{\frac{1}{2}}(3x-2)+\frac{2}{15}(x+1)^{\frac{3}{2}}\cdot 3$$

$$= \frac{1}{5}(x+1)^{\frac{1}{2}}\left[(3x-2)+2(x+1)\right]$$

$$= \frac{1}{5}(x+1)^{\frac{1}{2}}\cdot 5x = x\cdot(x+1)^{\frac{1}{2}}$$ which matches the integrand.

u-substitution: $u=x+1$, $du=dx$, $x=u-1$

$$\int (u-1)\cdot\sqrt{u}\,du = \int u^{\frac{3}{2}}-u^{\frac{1}{2}}\,du = \frac{2}{5}u^{\frac{5}{2}}-\frac{2}{3}u^{\frac{3}{2}}+C$$

$$= \frac{2}{5}(x+1)^{\frac{5}{2}}-\frac{2}{3}(x+1)^{\frac{3}{2}}+C$$

258. $\int x\sqrt{4x^2+9}\,dx = \frac{1}{12}\left(4x^2+9\right)^{3/2}+C; u=4x^2+9$

Check by using differentiation:

$$\frac{d}{dx}\left(\frac{1}{12}(4x^2+9)^{\frac{3}{2}}+C\right) = \frac{1}{12}\cdot\frac{3}{2}(4x^2+9)^{\frac{1}{2}}\cdot 8x = \frac{24}{24}x(4x^2+9)^{\frac{1}{2}} = x(4x^2+9)^{\frac{1}{2}}$$

which matches the integrand.

u-substitution: $u=4x^2+9$, $du=8x\,dx$, $\frac{du}{8}=x\,dx$

$$\int x\sqrt{4x^2+9}\,dx = \int \sqrt{u}\cdot\frac{du}{8} = \frac{2}{3}\frac{u^{\frac{3}{2}}}{8\cdot 4}+C = \frac{1}{12}(4x^2+9)^{\frac{3}{2}}+C$$

In the following exercises, find the antiderivative using the indicated substitution.

261. $\int (x+1)^4\,dx; u=x+1$; $du=dx$

$$\int (x+1)^4\,dx = \int u^4\,du = \frac{u^5}{5}+C = \frac{(x+1)^5}{5}+C$$

265. $\int \frac{x}{\sqrt{x^2+1}}\,dx; u=x^2+1$; $du=2x\,dx$, $\frac{du}{2}=x\,dx$

$$\int \frac{x\,dx}{\sqrt{x^2+1}} = \int \frac{1}{\sqrt{u}}\cdot\frac{du}{2} = \frac{1}{2}\frac{u^{\frac{1}{2}}}{2}+C = \frac{u^{\frac{1}{2}}}{4}+C = \frac{(x^2+1)^{\frac{1}{2}}}{4}+C$$

In the following exercises, use a suitable change of variables to determine the indefinite integral.

271. $\int x(1-x)^{99}\,dx$ $u=1-x$, $du=-dx$, $-du=dx$

$x=1-u$

$=\int u^{99}\cdot(1-u)\cdot(-du)=\int u^{99}\cdot(u-1)\,du=\int u^{100}-u^{99}\,du=\frac{u^{101}}{101}-\frac{u^{100}}{100}+C$

$=\frac{(1-x)^{101}}{101}-\frac{(1-x)^{100}}{100}+C$

273. $\int (11x-7)^{-3}\,dx$ $u=11x-7$, $du=11\,dx$, $\frac{du}{11}=dx$

$=\int u^{-3}\cdot\frac{du}{11}=\frac{u^{-2}}{-2\cdot 11}+C=\frac{(11x-7)^{-2}}{-22}+C$

275. $\int \cos^3\theta\sin\theta\,d\theta$ $u=\cos\theta$, $du=-\sin\theta\,d\theta$, $-du=\sin\theta\,d\theta$

$=\int u^3\cdot(-du)=\int -u^3\,du=-\frac{u^4}{4}+C=-\frac{\cos^4\theta}{4}+C$

276. $\int \sin^7\theta\cos\theta\,d\theta$ $u=\sin\theta$, $du=\cos\theta\,d\theta$

$=\int u^7\,du=\frac{u^8}{8}+C=\frac{\sin^8\theta}{8}+C$

In the following exercises, use a change of variables to evaluate the definite integral.

292. $\int_0^1 x\sqrt{1-x^2}\,dx$ $u=u(x)=1-x^2$; $du=-2x\,dx$, $\frac{du}{-2}=x\,dx$

$=\int_{u(0)}^{u(1)}\sqrt{u}\cdot\left(\frac{du}{-2}\right)=\int_1^0\frac{\sqrt{u}}{-2}\,du=\int_0^1\frac{\sqrt{u}}{2}\,du=\left[\frac{2}{3}\cdot\frac{u^{\frac{3}{2}}}{2}\right]_0^1$

$=\left[\frac{1}{3}(1)^{\frac{3}{2}}-\frac{1}{3}(0)^{\frac{3}{2}}\right]=\frac{1}{3}$

293. $\int_0^1 \frac{x}{\sqrt{1+x^2}}\,dx$ $u=u(x)=1+x^2$, $du=2x\,dx$, $\frac{du}{2}=x\,dx$

$=\int_{u(0)}^{u(1)}\frac{1}{\sqrt{u}}\cdot\frac{du}{2}=\left[\frac{1}{2}\cdot 2u^{\frac{1}{2}}\right]_1^2=\sqrt{2}-1$